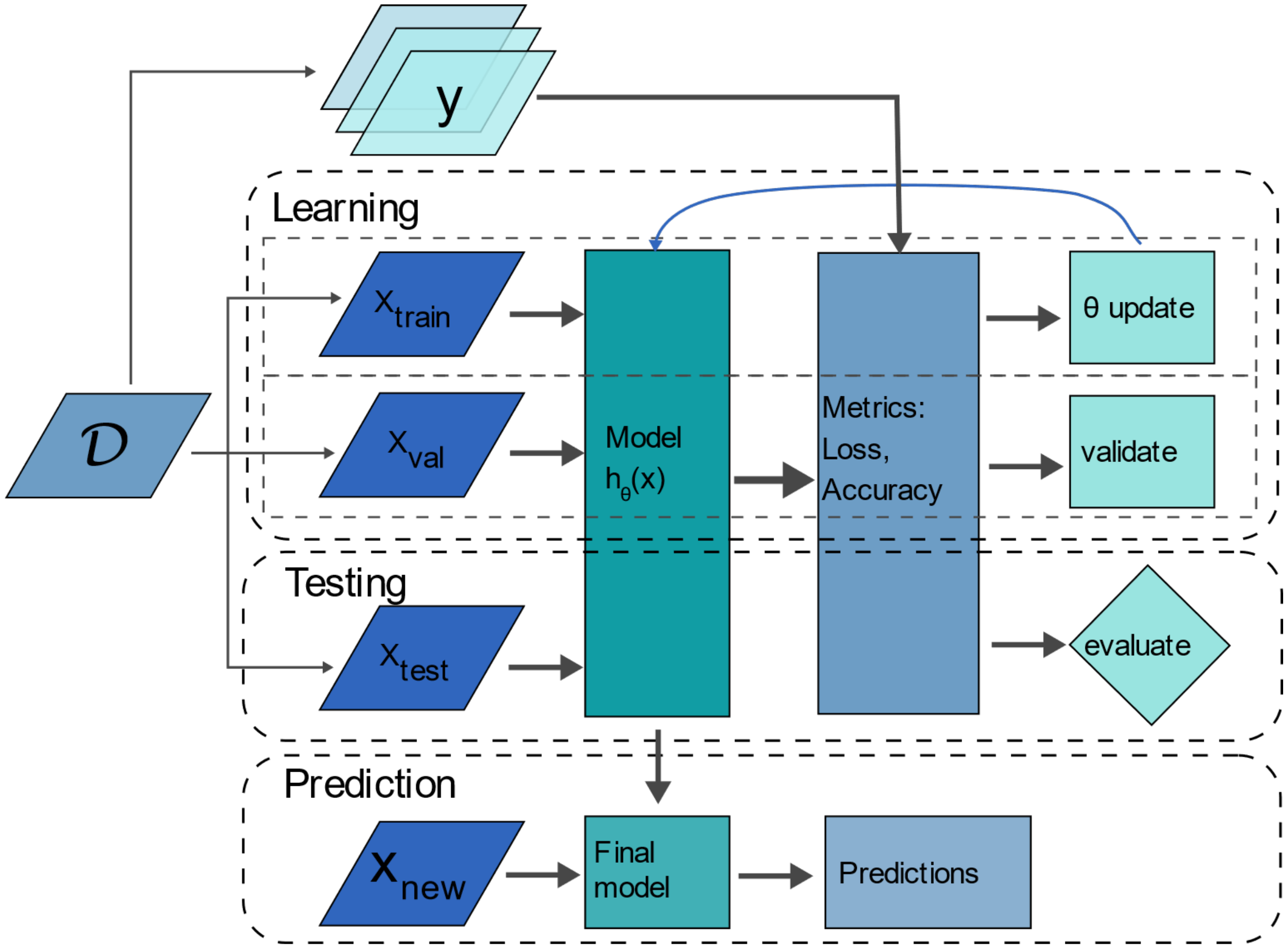
EEE4114F

Supervised Learning

Supervised Learning

- Given some labelled dataset $\mathcal{D} = \{(\mathbf{x}^{[i]}, \mathbf{y}^{[i]}), \ldots, (\mathbf{x}^{[n]}, \mathbf{y}^{[n]})\}$
- The goal is to learn a hypothesis function $h_\theta(\mathbf{x}) = \mathbf{y}$
- The subscript θ indicates that these are parameters to be learned
- Find values for θ that minimize/maximize a chosen objective function
- That objective function gives an indication to the learning algorithm how well it is predicting the ground truth label.

Supervised Learning Process



Next up:

Simple Linear Regression